

Metric on Vector Space

Definition

Motivation: We want to define “closeness / distance” of two vectors in a vector space.

Let V be a vector space, let $x, y \in V$ then

$$\text{dist}(x, y) = \text{dist}(x - y, y - y) = \text{dist}(x - y, 0)$$

To define distance, we only need to define the length of vectors. Let $x \in V$, we denote that $\|x\|$ be the length of x . Then $\|x\|$ should satisfy:

1. $\|x\| \geq 0$ (length is non-negative)
2. $\|x\| = 0$ if and only if $x = 0$ (length is zero only for zero vector)
3. $\|\alpha x\| = |\alpha| \|x\|$ (length is scaled by absolute value of scalar)
4. $\|x + y\| \leq \|x\| + \|y\|$ (triangle inequality)

Definition: A function $\|\cdot\| : V \rightarrow \mathbb{R}$ is called a norm on V if it satisfies the above properties.

Examples

Example 1: $\mathbb{R}$ is a vector space over $\mathbb{R}$. $\|x\| = |x|$ defines a norm on $\mathbb{R}$.

- $\|x\| = \frac{1}{2} |x|$ is also a norm on $\mathbb{R}$
- $\vdots$
- $\|x\| = c |x|, c > 0$ is also a norm on $\mathbb{R}$

There are infinitely many norms on the same vector space

Example 2: $\mathbb{R}^n$ is a vector space:

- Euclidean norm (2-norm):

$$\|x\|_2 = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}$$

Prove $\|\cdot\|_2$ is a norm on $\mathbb{R}^n$.

Proof:

1. $\|x\|_2 \geq 0$ is trivial.
2. $\|x\|_2 = 0 \Leftrightarrow \sum_{i=1}^n x_i^2 = 0 \Leftrightarrow x_i = 0 \quad \forall i = 1, 2, \dots, n.$
3. $\|\alpha x\|_2 = \sqrt{(\alpha x_1)^2 + (\alpha x_2)^2 + \dots + (\alpha x_n)^2} = |\alpha| \sqrt{x_1^2 + x_2^2 + \dots + x_n^2} = |\alpha| \|x\|_2.$

■

p -norm

$$\|x\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{\frac{1}{p}}$$

$\|\cdot\|_p$ is a norm on $\mathbb{R}^n$ for if and only if $p \geq 1$.

when $p \rightarrow \infty$ then $\|x\|_p$ is still a norm:

$$\forall x \in \mathbb{R}^n, \|x\|_\infty = \max_{i=1,2,\dots,n} |x_i|$$

Property:

$$\|x\|_p \leq \|x\|_q \text{ if } p \geq q$$

For instance:

$$\|x\|_1 \geq \|x\|_2 \geq \|x\|_\infty$$

Example 3: $\mathbb{R}^{m \times n}$ is a vector space:

- $\mathbb{R}^{m \times n}$ can be viewed as $\mathbb{R}^{mn}$

$$\begin{pmatrix} x_{11} & x_{12} & \dots & x_{1n} \\ x_{21} & x_{22} & \dots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \dots & x_{mn} \end{pmatrix} \Rightarrow \begin{pmatrix} x_{11} \\ \vdots \\ x_{1n} \\ x_{21} \\ \vdots \\ x_{2n} \\ \vdots \\ x_{m1} \\ \vdots \\ x_{mn} \end{pmatrix}$$

We can define vector p -norm on $\mathbb{R}^{mn}$ for a matrix A on $\mathbb{R}^{m \times n}$:

$$\|A\|_p = \left(\sum_{i=1}^m \sum_{j=1}^n |x_{ij}|^p \right)^{\frac{1}{p}}$$

For instance when $p = 2$ then:

$$\|A\|_F = \|A\|_{2, \text{vec}} = \left(\sum_{i=1}^m \sum_{j=1}^n |x_{ij}|^2 \right)^{\frac{1}{2}}$$

which is called Frobenius norm. When $p = \infty$ then:

$$\|A\|_{\infty, \text{vec}} = \max_{i=1,2,\dots,m} \max_{j=1,2,\dots,n} |x_{ij}|$$

$\mathbb{R}^{m \times n}$ can also be viewed as linear transformations $\mathbb{R}^n \rightarrow \mathbb{R}^m$. We define matrix p -norm by:

$$\|A\|_p \stackrel{\text{def}}{=} \max_{x \in \mathbb{R}^n, x \neq 0} \frac{\|Ax\|_p}{\|x\|_p} = \max_{\|x\|_p = 1} \|Ax\|_p$$

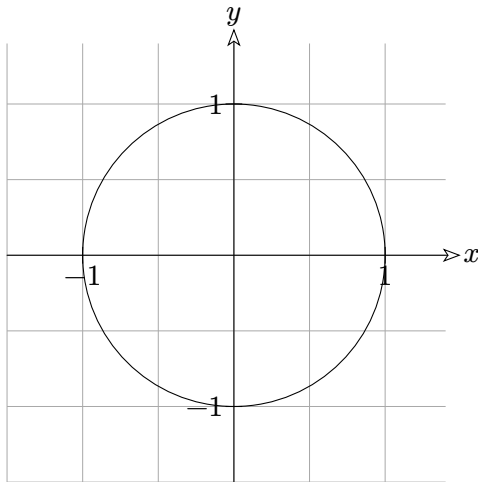


Figure 1: $\{x \mid x \in \mathbb{R}^n \text{ and } \|x\|_2 = 1\}$

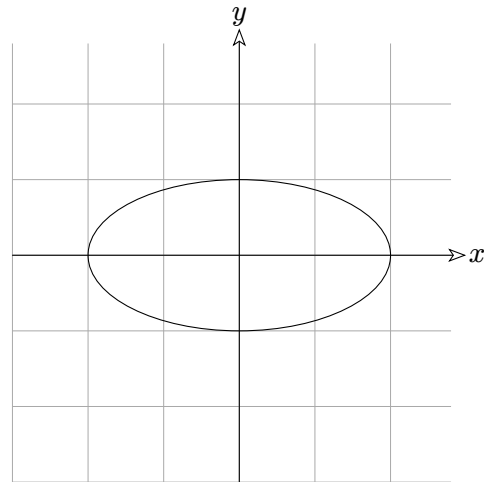


Figure 2: $\{Ax \mid A \in \mathbb{R}^{m \times n} \text{ and } \|x\|_2 = 1, x \in \mathbb{R}^n\}$

So that $\|A\|_2$ is just to find a point on the ellipse that has the largest distance from the origin.

In the analytical view:

$$\begin{aligned}\|A\|_2^2 &= \max_{\|x\|_2=1} \|Ax\|_2^2 \\ &= \max_{\|x\|_2=1} (x^T A^T A x) \\ &= \text{max eigenvalue of } A^T A\end{aligned}$$

then:

$$\begin{aligned}\|A\|_2 &= \sqrt{\text{max eigenvalue of } A^T A} \\ &\stackrel{\text{aka}}{=} \text{max singular value of } A\end{aligned}$$

$\|\cdot\|_2$ is also called the operator norm of A .

- When $p = 1$: (max column 1-norm)

$$\|A\|_1 = \max_{j=1, \dots, n} \left(\sum_{i=1}^m |x_{ij}| \right)$$

- When $p = \infty$: (max row 1-norm)

$$\|A\|_\infty = \max_{i=1, \dots, m} \left(\sum_{j=1}^n |x_{ij}| \right)$$

- The default norm of a matrix is $\|\cdot\|_2$
- We can have other matrix norms, e.g. we can use different norms in $\mathbb{R}^n$ (p -norm) and $\mathbb{R}^m$ (q -norm):

$$\|A\|_{p \rightarrow q} = \max_{\|x\|_p=1} \|Ax\|_q$$